

SUPPLEMENTARY MATERIAL TO COGNITIVE WARRANT RESEARCH

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This document supplements my research documents, providing background information, intermediate steps, or detailed explanations for those that are interested. It is a work in progress.

The concept of information is too broad to be captured completely by a single definition. However, for any probability distribution, we define a quantity called the *entropy*, which has many properties that agree with the intuitive notion of what a measure of information should be. This notion is extended to define *mutual information*, which is a measure of the amount of information one random variable contains about another. Entropy then becomes the self-information of a random variable. Mutual information is a special case of a more general quantity called *relative entropy*, which is a measure of the distance between two probability distributions. All these quantities are closely related and share a number of simple properties....

...these quantities arise as natural answers to a number of questions in communication, statistics, complexity, and gambling. That will be the ultimate test of the value of these definitions.

—Thomas M. Cover & Joy A. Thomas,
Elements of Information Theory (2006, p. 13)

CHAPTER 1: Probability and Information Theory

1.1. Conditional Probability

Fundamental terms of probability

- An **experiment** is any action or process whose outcome is subject to uncertainty.
- The **sample space** is the set of all possible outcomes of that experiment.
- An **event** is any collection (subset) of outcomes contained in the sample space.
- An event is **simple** if it consists of exactly one outcome and **compound** if it consists of more than one outcome.
- A and B are considered **independent** events if the occurrence or nonoccurrence of one event has no bearing on the chance that the other will occur.

We intuitively know that a probability is some measure of an event's likelihood, or the certainty of an outcome, or our degree of belief in a proposition or hypothesis. The probability of an event or outcome is 0, 1, or any number between.

If we have defined an event A , then $P(A)$ is the marginal, or unconditional, probability of event A occurring without consideration of any other events. $P(B)$ is thus the marginal probability of event B . If A and B are independent, as defined above, then knowing whether or not B has occurred, or been observed, provides no information about, or has no effect on, the subsequent (non)occurrence of A . If, however, these two events are *not* independent, then the occurrence of B does provide

information about, or has an effect on, the subsequent occurrence or observation of A . In this case, the probability of A is conditioned by B and is notated as $P(A | B)$.

Definition 1.1 (Conditional probability). Let A and B be two events with $P(B) \neq 0$. The conditional probability of A given B is defined to be

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

The notation $P(A \cap B)$ represents the probability of *both* events A and B occurring.

In the sense of conditional probability being how we update our beliefs in light of the evidence we observe, $P(A)$ is called the **prior probability** of A and $P(A | B)$ the **posterior probability** of A . The distinction is whether the probability is before being updated by evidence or after.

As an example, let A be the event that a person is wet at any moment and B the event that it is raining at that person's location. If we know that B has occurred or is occurring, the prior probability that this person is wet can be updated. In this situation, the person's state of being drenched is conditional upon B having occurred. We will see below that, on average, conditioning (i.e., knowing the state of one random variable) reduces some uncertainty about another random variable (but leaves that uncertainty unchanged if those random variables are independent). If I know the result of flipping one coin, that does not affect my uncertainty about the result of flipping a different coin; however, if I know that it is raining at a person's location, that reduces my uncertainty (to some degree) about whether he or she is wet.

1.2. Entropy

A **random variable** is a quantity whose value depends on the outcome of an uncertain process. The name is slightly misleading — it doesn’t mean arbitrary or haphazard. It means the value is determined by something not yet observed.¹ A coin flip, a person’s response time on a trial, and whether two sentences invoke the same word sense are all random variables: before you observe them, they could take more than one value; after you observe them, they take exactly one. The possible values of a **discrete random variable** are finite or “countably infinite.” An example of a discrete random variable is the sum of rolling two dice, as its possible values are the integers 2 through 12. What makes a random variable mathematically tractable is that even though individual outcomes are uncertain, the *distribution* of outcomes — which values are likely and which are unlikely — can be known and reasoned about precisely.

Entropy measures how uncertain that distribution is. More precisely, it quantifies how surprised you should expect to be by the outcome — or equivalently, how much information you gain when you learn it. A fair coin has high entropy: you genuinely don’t know what’s coming, so the outcome tells you something. A coin weighted to land heads 99% of the time has low entropy: you already know roughly what will happen, so the outcome tells you little. Crucially, entropy is a property of the *distribution*, not of any single observation and not of what the outcomes actually are.

You may be familiar with the term entropy from the second law of thermodynamics and some version of ‘over time, everything becomes more disordered.’ I

¹Indeed, Deisenroth et al. (2020) note that “the name ‘random variable’ is a great source of misunderstanding as it is neither random nor is it a variable. It is a function.”

won't discuss the use of this term in thermodynamics versus information theory, but perhaps associating disorder with uncertainty will make it easier to understand what is meant by entropy here.

Definition 1.2 (Entropy). Formally, the entropy of the random variable X is

$$H(X) = - \sum_x p(x) \log p(x). \quad (1.1)$$

For each of the possible values of the random variable, we are multiplying its probability $p(x)$ by the logarithm of its probability $\log p(x)$. The entropy $H(X)$ is the negated sum of these products.

Earlier, I said that entropy quantifies how surprised you should expect to be by the outcome of an uncertain process. Surprise is the result of an unexpected event, or, in other words, an event with a low probability. Thus, *surprise and likelihood are inversely related*: the greater the likelihood of an event, the less surprise when it occurs.

A property of logarithms is $\log M^k = k \cdot \log M$. Because of the distributive property, we can bring the negative sign into our summation, and Equation (1.1) can be written as

$$\begin{aligned} H(X) &= \sum_x -p(x) \log p(x) \\ &= \sum_x p(x) \cdot (-1) \cdot \log p(x) \\ &= \sum_x p(x) \cdot \log(p(x))^{-1} \\ &= \sum_x p(x) \cdot \log \frac{1}{p(x)} \end{aligned}$$

The last term $\frac{1}{p(x)}$ is the inverse of the likelihood of x and thus encodes surprisal in entropy. Taking the logarithm transforms the scale but not the inverse relationship. We can now say that for each possible value of X , we are multiplying the surprise associated with that value by the value's likelihood. Entropy is the sum of these products. You may be familiar with a weighted sum. The expected value of a random variable is the sum of its possible values each weighted by their probability. Entropy is thus the expected amount (or weighted average) of surprise over the distribution of the random variable.

An immediate consequence of Definition 1.2 is $H(X) \geq 0$: entropy is always non-negative. If the outcome is certain, $H(X) = 0$; there is nothing to learn. Otherwise $H(X) > 0$; some uncertainty always remains.

Figure 1.1 shows entropy for the simplest possible case: a Bernoulli variable that takes only two values (such as 0 or 1), with probability p for one outcome and $(1 - p)$ for the other. Entropy is zero at the extremes — when the outcome is a near-certainty in either direction ($p = 0$ or $p = 1$) — and peaks at $p = \frac{1}{2}$, when both outcomes are equally likely and uncertainty is at its maximum. The shape of this figure is important in the “H3 as a Mathematical Consequence of the CI Framework” section of my research document.

1.3. Conditional Entropy

Conditional entropy measures how much uncertainty remains about one random variable once the value of another is known, averaged over the values that other variable can take.² If knowing X tells you a lot about Y , the conditional entropy

²This and following sections are based primarily on *Elements of Information Theory* (2nd ed.) by Cover and Thomas.

Binary Entropy Function

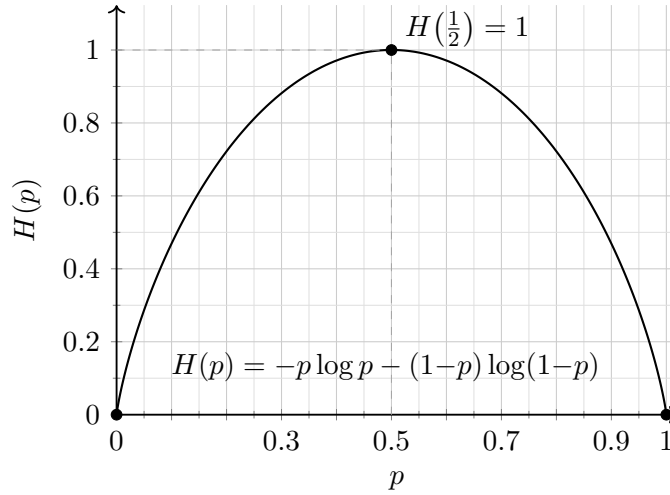


Figure 1.1: Entropy of a two-outcome variable as a function of the probability p of one outcome. Uncertainty — and therefore entropy — is zero when the result is a near-certainty and is maximized when both outcomes are equally likely.

$H(Y | X)$ is low — little uncertainty is left. If knowing X tells you nothing about Y , $H(Y | X)$ equals the original entropy of Y — knowledge of X bought you nothing.

The definition computes this by asking: for each possible value of X , how uncertain is Y ? It then averages those uncertainties, weighted by how often each value of X actually occurs:

$$H(Y | X) = \sum_x p(x) H(Y | X=x) \quad (1.2)$$

$$= - \sum_x p(x) \sum_y p(y | x) \log p(y | x) \quad (1.3)$$

$$= - \sum_x \sum_y p(x, y) \log p(y | x). \quad (1.4)$$

where $p(x, y)$ is the joint distribution.

In the Theoretical Framework chapter of my research document, I introduce the binary random variable Y indicating whether a word used in two different contexts instantiates the same sense of the word, and the continuous quantity semantic distance, denoted d , representing the “dissimilarity” of the two contexts. The conditional probability $P(Y \mid d)$ thus defines a mapping from semantic distance to sense identity. At low distances the mapping is highly predictable toward same-sense identity; at high distances, it is equally predictable toward different-sense identity. At intermediate distances the mapping becomes uncertain, which can be formalized using **conditional entropy**:

$$H(Y \mid D=d) = - \sum_{y \in \{0,1\}} p(y \mid d) \log p(y \mid d) \quad (1.5)$$

You may have noted an apparent discrepancy between Equation (1.5) and Equation (1.3): there is no weighting by how often each value of d occurs. The resolution is that the two equations denote different objects. Equation (1.3) is the **averaged** conditional entropy $H(Y \mid D)$: a single number, obtained by averaging the residual uncertainty in Y across the entire distribution of distances. Equation (1.5) is the **pointwise** conditional entropy at one particular distance, written here as $H(Y \mid D=d)$. It is not a number but a function of d — the entropy of a single conditional distribution. No weighting appears because no averaging has been performed. The two are related by an expectation over the distribution of distances, which for continuous d with density $f(d)$ is

$$H(Y \mid D) = \mathbb{E}_D[H(Y \mid D=d)] = \int f(d) H(Y \mid D=d) dd. \quad (1.6)$$

It is tempting to attribute the difference instead to the fact that semantic distance functions as a controlled independent variable in my experimental design rather

than as a “random” variable in the sense discussed above. That framing does not account for the missing weights. A controlled variable still has an empirical design distribution over which one could average, and conversely one can always condition on a single realized value of a genuinely random variable. The operative distinction is between conditioning on a particular value and averaging across values — not between fixed and random quantities.

The pointwise form is the one the framework requires. Cognitive information is defined at each distance, $CI(d) \equiv I(Y; S \mid d)$, and Hypothesis 3 concerns the *shape* of that function across the range of d — a claim that presupposes a quantity varying with distance rather than a scalar summary. The aggregate is recovered when needed by the same expectation used in Equation (1.6). Throughout, I write $H(Y \mid D=d)$ when conditioning on a particular value and $H(Y \mid D)$ for the average.

$H(Y \mid D=d)$ represents the disambiguation load. My claim is that cognitive processing difficulty in the primed sensibility judgment task of Experiment 1 increases with this uncertainty: higher $H(Y \mid d)$ corresponds to longer response times as the cognitive system labors to resolve which sense is intended.

Conditioning reduces entropy. $H(X \mid Y) \leq H(X)$, with equality if and only if X and Y are independent. Knowing Y can only reduce — or leave unchanged — your uncertainty about X . This is only true on average, not for every individual observation. In a court case, a specific piece of new evidence might increase uncertainty by opening up possibilities that hadn’t been considered. But across all possible pieces of evidence, the expected effect is to reduce uncertainty, not increase it.

Note. The order of conditional entropy matters: knowing X does not generally reduce uncertainty about Y by the same amount that knowing Y reduces uncertainty

about X , so $H(Y | X) \neq H(X | Y)$ in general. However, the *difference* each variable makes is symmetric: $H(X) - H(X | Y) = H(Y) - H(Y | X)$. Both expressions equal the mutual information between X and Y — the uncertainty each one resolves about the other — introduced in Section 1.5.

1.4. Relative Entropy

Information theory provides a measure of the distance between two distributions.

Definition 1.3. The relative entropy or Kullback-Leibler (KL) divergence between two probability mass functions $p(x)$ and $q(x)$ is defined as

$$D(p \parallel q) = \sum_x p(x) \log \frac{p(x)}{q(x)} \quad (1.7)$$

We will use the conventions that $0 \log \frac{0}{0} = 0$, $0 \log \frac{0}{q} = 0$, and $p \log \frac{p}{0} = \infty$.

There are several interpretations of this concept that can be useful later. The original motivation was as an expected log likelihood ratio as a way to distinguish between two distributions, i.e. a measure of how much an approximating probability distribution Q is different from the true distribution P .

Another interpretation is as the information gain achieved if the true distribution P would be used instead of the currently used Q ; conversely, it can be interpreted as the amount of information lost when Q is used to approximate P . Put in terms of Bayesian inference, relative entropy is a measure of the information gained by revising one's beliefs from the prior probability distribution Q to the posterior probability distribution P . We will be using this last interpretation in our data analysis.

Note that relative entropy is not a true distance between distributions since it is not symmetric ($D(p \parallel q) \neq D(q \parallel p)$ in general) and does not satisfy the triangle inequality (including a third distribution u , $D(p \parallel q) \leq D(p \parallel u) + D(u \parallel q)$ does not hold in general). Although relative entropy is technically a “divergence” and not a distance, it is often useful to think of it as a “distance” between distributions.

1.5. Mutual Information

Mutual information $I(X; Y)$ measures how much knowing one variable tells you about another. Equivalently, it is the reduction in uncertainty about X that results from learning Y — or vice versa, since the relationship is symmetric. Two variables that are completely unrelated have zero mutual information; the more tightly their values are linked, the higher it is.

$$I(X; Y) = D(p(x, y) \parallel p(x)p(y)) \quad (1.8)$$

$$= \sum_x \sum_y p(x, y) \log \frac{p(x, y)}{p(x)p(y)}. \quad (1.9)$$

The formula compares the actual joint distribution of X and Y — how they behave together — to what their joint distribution *would* be if they were completely independent. When the two distributions match, i.e. the random variables *are* independent, the variables share no information and $I(X; Y) = 0$. The more the true and independent distributions diverge, the more the variables mutually provide information about each other.

1.6. Relationship Between Entropy and Mutual Information

Entropy, conditional entropy, and mutual information are not separate concepts — they are different ways of partitioning the same underlying uncertainty (or poten-

tial information). The key identity is:

$$I(X; Y) = H(X) - H(X | Y). \quad (1.10)$$

In words: mutual information is exactly the uncertainty about X that disappears once you know Y . Or, rearranging Equation 1.10,

$$H(X) = I(X; Y) + H(X | Y). \quad (1.11)$$

Everything that was uncertain about X either (a) gets resolved by learning Y , which is the mutual information, or (b) remains uncertain even after learning Y , which is the conditional entropy. Nothing is double-counted; nothing is lost.

Because mutual information is symmetric — X tells you as much about Y as Y tells you about X — the same decomposition runs in both directions:

$$I(X; Y) = H(Y) - H(Y | X). \quad (1.12)$$

A third way to express the same relationship uses joint entropy $H(X, Y)$, the total uncertainty when X and Y are considered together:

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \quad (1.13)$$

If X and Y were independent, their joint uncertainty would equal the sum of their individual uncertainties. Mutual information is the amount by which the joint uncertainty falls *short* of that sum — the overlap that would be double-counted if the marginals were naively added (see Figure 1.2).

Collecting these results:

$$I(X; Y) = H(X) - H(X | Y) \quad (1.14)$$

$$= H(Y) - H(Y | X) \quad (1.15)$$

$$= H(X) + H(Y) - H(X, Y) \quad (1.16)$$

$$= I(Y; X) \quad (1.17)$$

$$I(X; X) = H(X) \quad (1.18)$$

The last line has an intuitive reading: a variable contains exactly one variable's worth of information about itself — its own full entropy, no more and no less — which is why this document's epigraph referred to entropy as self-information.

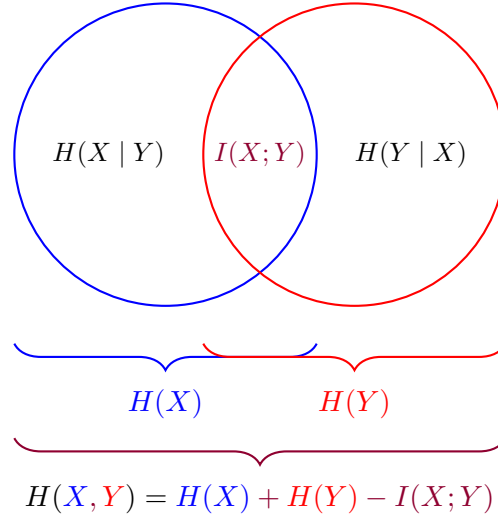


Figure 1.2: Entropy and mutual information as overlapping regions of $H(X)$ and $H(Y)$. $I(X; Y)$ is the shared overlap — uncertainty that either variable resolves about the other. The conditional entropies $H(X | Y)$ and $H(Y | X)$ are the portions exclusive to each variable. The joint entropy $H(X, Y)$ is the total area of both circles combined (*without counting the mutual information twice*).

1.7. Conditional Mutual Information

In the previous section, we covered the relationships among entropy, conditional entropy, and mutual information (see, for example, Equation 1.11). This naturally leads us to the concept of conditional mutual information.

Conditional mutual information $I(X; Y | Z)$ measures how much knowing X reduces uncertainty about Y once Z is already known:

$$I(X; Y | Z) = H(Y | Z) - H(Y | X, Z) \quad (1.19)$$

Equivalently, it is the mutual information between X and Y computed within each stratum of Z , then averaged over the distribution of Z :

$$I(X; Y | Z) = \mathbb{E}_Z[I(X; Y | Z = z)] = \sum_z P(z) I(X; Y | Z = z), \quad (1.20)$$

where the sum specializes the expectation to discrete Z ; for continuous Z it becomes an integral against the density. Like marginal mutual information, $I(X; Y | Z) \geq 0$ always, with equality if and only if $(X \perp Y) | Z$ (i.e., X and Y are conditionally independent given Z).

A Side Note

What does conditional independence mean? The cleanest definition is a factorization: for all x, y and all z with $P(z) > 0$,

$$(X \perp Y) \mid Z \iff P(x, y \mid z) = P(x \mid z) P(y \mid z). \quad (1.21)$$

Wherever the conditional probabilities are defined, this is equivalent to the form that reads more naturally, $P(x \mid y, z) = P(x \mid z)$. In words: once Z is known, Y tells you nothing further about X .

A more roundabout way to say the same thing: within any fixed stratum $Z = z$, the distribution of X is identical for every value of Y , and the distribution of Y is identical for every value of X . The symmetry here comes from the factorization itself, which treats X and Y interchangeably — conditional independence is a relation *between* two variables, not something one variable has with respect to another.

The really interesting part (to me) is that $X \perp Y$ neither implies nor is implied by $(X \perp Y) \mid Z$. How do two dependent variables become independent once a third is known — and, stranger, how do two independent variables become dependent? Conditioning on Z can either *decrease* or *increase* the dependence between X and Y , and the two directions correspond to recognizable causal structures.

Conditioning decreases dependence, $I(X; Y \mid Z) < I(X; Y)$, when the association between X and Y runs through Z . Two distinct structures do this: Z is a *mediator* on a chain $X \rightarrow Z \rightarrow Y$, or Z is a *common cause* $X \leftarrow Z \rightarrow Y$. Either way, knowing Z explains the association away; if the depicted path is the only one connecting X and Y , it vanishes entirely and $I(X; Y \mid Z) = 0$.

Conditioning increases dependence, $I(X; Y | Z) > I(X; Y)$, when Z is a *common effect* — a “collider,” $X \rightarrow Z \leftarrow Y$. Conditioning on a common effect induces dependence between causes that were marginally independent, or amplifies dependence that was already there. This is the direction that tends to strain intuition, so it is worth seeing at its most extreme.

Example

Let X and Y be independent fair coin flips, each coded 0 or 1, and let $Z = X \oplus Y$ indicate whether the two flips agreed (equivalently, whether they disagreed). Marginally, X tells you nothing about Y : $I(X; Y) = 0$. But condition on Z . If you know the flips disagreed, then learning $X = 0$ fixes $Y = 1$ with certainty; learning $X = 1$ fixes $Y = 0$. The same holds, with the values swapped, if you know they agreed. Within either stratum of Z , X determines Y completely, so $I(X; Y | Z) = 1$ bit — the maximum possible for a binary variable.

Nothing about X or Y changed. What changed is that Z is a joint function of both, so knowing its value converts any fact about one into a fact about the other. Dependence induced by conditioning on a common effect is a property of the observer’s information state, not of the underlying variables.

One consequence worth stating explicitly: although $I(X; Y)$ and $I(X; Y | Z)$ are each nonnegative, their difference $I(X; Y) - I(X; Y | Z)$ — the interaction information — has no sign restriction. The two cases above are precisely what that unrestricted sign encodes. And in real data, where mediating, confounding, and collider paths generally coexist, the direction you observe reflects their net effect rather than any single structure.

1.7.1. Cognitive Warrant as Uncertainty Reduction

Definition 1.4. In my Theoretical Framework chapter, I define the binary variable $S \in \{0, 1\}$ as the **heading congruence** indicator: $S = 1$ when two contexts are assigned to the same LC subject heading (**congruent**), and $S = 0$ when they are assigned to different LC subject headings (**incongruent**). Neither value is the marked or active condition — congruence and incongruence are symmetric categorical outcomes of the heading assignment process, distinguished by whether the two contexts fall within or across an LCSH authority record boundary. When it is necessary to refer to the aggregate pattern of heading congruence values across items or distance bins, I will use **categorical congruence structure**.

This yields a refined conditional distribution with associated entropy $H(Y \mid d, S)$. The **cognitive information** provided by categorical congruence structure is defined as:

$$CI(d) = H(Y \mid d) - H(Y \mid d, S)$$

Formally, $CI(d)$ quantifies, conditional on semantic distance d , the reduction in uncertainty about sense identity Y afforded by heading congruence S — that is, by (explicitly or implicitly) knowing whether the two contexts are (congruently) assigned to the same subject heading or (incongruently) assigned to different headings.

The formulation of cognitive information can be expanded to show the significance of conditional probabilities in its calculation.

$$CI(d) = - \underbrace{\sum_{y \in \{0,1\}} p(y \mid d) \log p(y \mid d)}_{H(Y \mid d)} + \underbrace{\sum_{s \in \{0,1\}} p(s \mid d) \sum_{y \in \{0,1\}} p(y \mid d, s) \log p(y \mid d, s)}_{-H(Y \mid d, S)}$$

Categorical congruence structure is informational to the extent that it reduces processing time by resolving ambiguity. Cognitive warrant is therefore reinterpreted as: *the extent to which a knowledge organization system reduces cognitive processing difficulty by resolving uncertainty in semantic interpretation.*

Cognitive information can be recognized as conditional mutual information:

$$\text{CI}(d) = H(Y \mid d) - H(Y \mid d, S) \equiv I(Y; S \mid d) \quad (1.22)$$

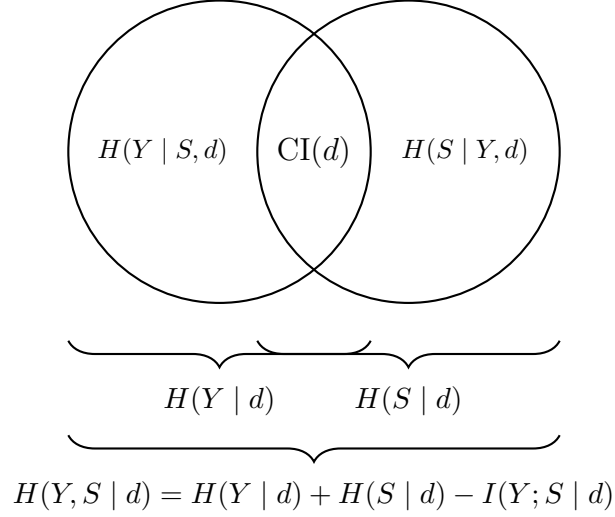
Having this form means that every theorem proven about $I(X; Y \mid Z)$ applies directly to $\text{CI}(d)$, and every informational identity translates into a claim about cognitive warrant. This is also the master key to the derivations such as warrant efficiency (η) and pointwise cognitive information (pCI).

Two corollaries are immediate. First, because conditional mutual information is always non-negative, $\text{CI}(d) \geq 0$ unconditionally: LCSH categorical structure cannot, *on average*, increase uncertainty about sense identity at any given distance. Second, the chain rule applied to the three-variable system (Y, d, S) yields:

$$\begin{aligned} I(Y; d, S) &= H(Y) - H(Y \mid d, S) \\ &= H(Y) - [H(Y \mid d) - I(Y; S \mid d)] \\ &= [H(Y) - H(Y \mid d)] + I(Y; S \mid d) \\ &= \underbrace{I(Y; d)}_{\text{continuous contribution}} + \underbrace{I(Y; S \mid d)}_{\text{CI}(d): \text{ categorical contribution}} \end{aligned}$$

Total information about sense identity from the complete model decomposes into the contribution of continuous distance alone plus the additional contribution of the

categorical congruence structure conditional on distance. The alignment with Trott and Bergen’s (2023) hybrid meaning theory is clear.



$$\Rightarrow 0 \leq \text{CI}(d) = H(Y | d) + H(S | d) - H(Y, S | d) \leq \min(H(Y | d), H(S | d))$$

Figure 1.3: Entropy and mutual information (all conditioned on d) as overlapping regions of $H(Y | d)$ and $H(S | d)$. $\text{CI}(d) \equiv I(Y; S | d)$ is the shared overlap — uncertainty that either variable resolves about the other, conditioned on d . The conditional entropies $H(Y | S, d)$ and $H(S | Y, d)$ are the portions exclusive to each variable. The conditional joint entropy $H(Y, S | d)$ is the total area of both circles combined. From the figure, we can informally conclude that $\text{CI}(d)$ is non-negative and no greater than either $H(Y | d)$ or $H(S | d)$. This will be important for the construct of warrant efficiency and the redundancy bound.

1.8. Summary

This chapter provided explanations of probability and information theory concepts found in my cognitive warrant research. Reviewed topics include marginal and conditional probability, entropy of a random variable, conditional entropy, relative entropy or KL divergence, mutual information, and conditional mutual information.

The chapter's aim was not exhaustive coverage of these topics. The hope is that this supplementary material has provided the minimum working fluency required to follow the argument and enable genuine engagement with the research by a motivated reader.